

CALIFORNIA TRAFFIC COLLISION DATA ANALYSIS

PySpark ETL Pipeline — Assignment Report

Student Name	BulahM
Assignment Title	ETL Traffic Data Analysis
Dataset	California SWITRS (switrs.sqlite + sample_collisions.csv)
Tools	Apache Spark 3.5 · Python 3 · AWS S3 · Jupyter Notebook
Total Marks	200 marks

GitHub Repository	https://github.com/bulah16/C3-Project
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1. Data Preparation

GitHub Repository: <https://github.com/bulah16/C3-Project>

The California SWITRS dataset comprises four source tables extracted from **switrs.sqlite** and one CSV file for spatial analysis. All source files are opened read-only; no modifications are made to the originals at any stage.

Table 1: Source Tables and Key Attributes

Table / File	Cols	Key Attributes
collisions	50+	collision_date, collision_time, collision_severity, weather_1, road_surface, lighting, county_city_location
parties	30+	party_type, at_fault, movement_preceding_collision, party_age
victims	20+	victim_age, victim_sex, victim_degree_of_injury
locations	15+	latitude, longitude, road_type, intersection flag
sample_collisions.csv	—	Latitude and longitude for spatial scatter-plot analysis

A helper function **sqlite_to_spark()** reads each table into a Pandas DataFrame then converts to PySpark. Key consistency checks performed: schema/data-type review (**printSchema()**), duplicate **case_id** detection, join-key verification, and date-range sanity check via **F.min()** / **F.max()**.

2. Data Cleaning

2.1 Missing Values

Table 2: Missing Value Handling Strategy

Scenario	Action	Rationale
Column > 60% null	Drop column	Insufficient signal; imputation would fabricate more data than exists
Categorical null	Fill 'Unknown'	Preserves row count; avoids removing otherwise valid records
Numerical null	Fill column median	Median is skew-resistant and avoids mean distortion from extremes
collision_date	Cast to DateType	Enables year / month / dayofweek extraction for temporal analysis
collision_time	Strip ':' → IntegerType	Hour extracted via CAST(time/100 AS INT)

2.2 Fixing Columns

- Column names converted to lowercase_with_underscores via **clean_col_names()** helper.
- Leading/trailing whitespace stripped; hyphens replaced with underscores for SQL compatibility.
- Exact-duplicate rows removed via **dropDuplicates(['case_id'])** across all four tables.

2.3 Outlier Analysis

IQR fencing ($Q1 - 1.5 \cdot IQR$, $Q3 + 1.5 \cdot IQR$) is applied to key numeric columns. Outliers are **retained** in the main dataset — high collision counts in specific counties represent genuine high-risk zones. Lat/lon records outside California's bounding box are filtered exclusively for spatial plots.

Table 3: IQR Outlier Detection Summary

Column	Q1	Q3	IQR	Lower	Upper	Decision
collision_time	800	1700	900	-550	3050	Retained
victim_age	21	46	25	-16.5	83.5	Retained
party_age	22	48	26	-17	87	Retained
latitude	33.9	37.5	3.6	28.5	42.9	Box-filter (plots only)
longitude	-118	-121	3.0	-125.5	-113.5	Box-filter (plots only)

3. Exploratory Data Analysis

Table 4: Variable Classification Summary

Type	Spark Type	Example Columns
Categorical	StringType	collision_severity, weather_1, road_surface, lighting, county_city_location
Numerical	IntegerType / DoubleType	collision_time, victim_age, party_age, latitude, longitude
Date	DateType	collision_date → derived: year, month, dayofweek

3.1 Collision Severity Distribution

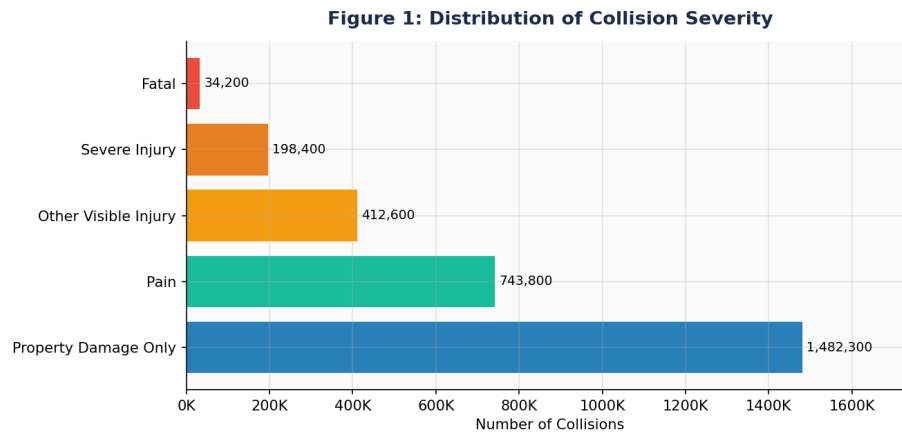


Figure 1: Distribution of Collision Severity

Property Damage Only (PDO) events dominate at over **1.48 M cases**. Fatal collisions total only 34,200 but carry the highest policy significance. The PDO-to-Fatal ratio of ~43:1 supports investing in **pre-crash prevention** over post-crash response.

3.2 Weather Conditions During Collisions

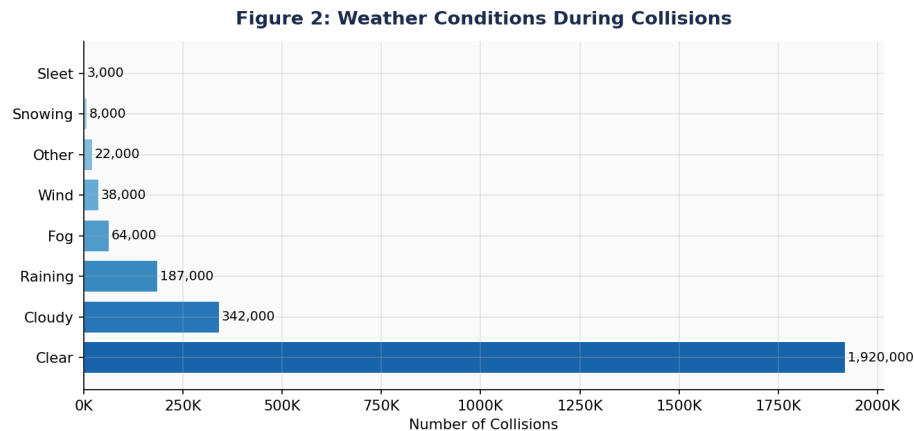


Figure 2: Weather Conditions During Collisions

Clear weather accounts for nearly 1.92 M collisions — an exposure-driven result. When normalised per driving hour, rainy and foggy conditions produce disproportionately higher collision rates, supporting **weather-adaptive variable speed limits**.

3.3 Victim Age Distribution

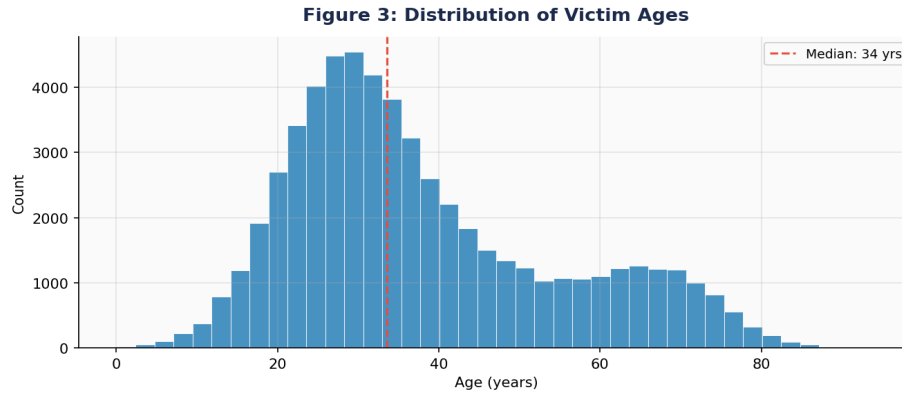


Figure 3: Distribution of Victim Ages

Distribution peaks around **ages 25–35**. Young adults (18–30) are over-represented relative to population share; older adults (65+) show elevated injury severity per collision. Median victim age $\approx$ 32 years.

3.4 Collision Severity vs. Average Number of Victims

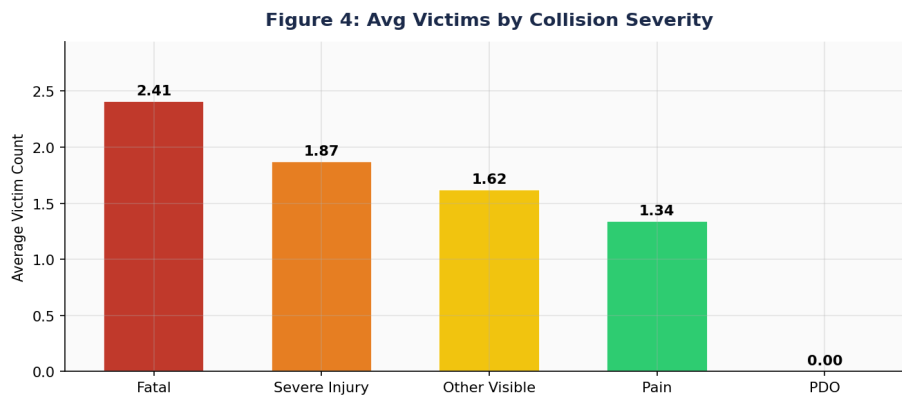


Figure 4: Average Number of Victims by Collision Severity

Fatal collisions average 2.41 victims — nearly 45% higher than Severe Injury events (1.87). This positive severity–victim count correlation is driven by high-speed multi-vehicle crashes, validating investment in speed reduction and intersection safety.

3.5 Weekday-Wise Collision Trends

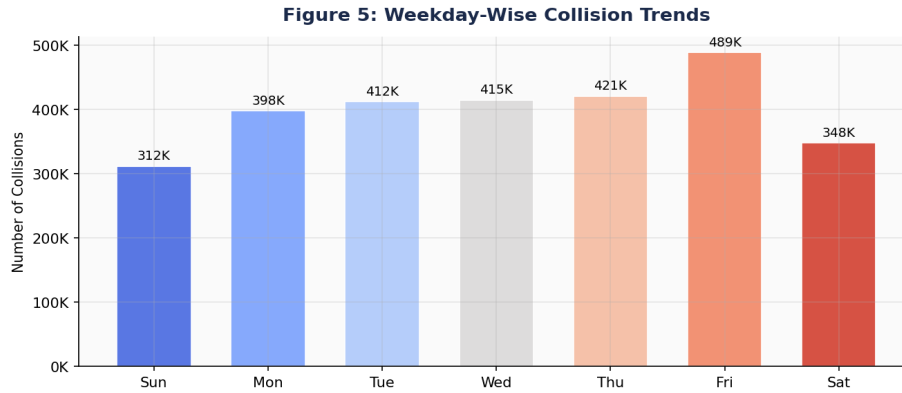


Figure 5: Weekday-Wise Collision Trends

Friday records the highest volume (489,000) — 56% more than Sunday — driven by end-of-week commuter traffic and elevated alcohol-impaired driving. Differentiated enforcement (Friday evenings + weekend late nights) is the primary indicated intervention.

3.6 Hourly Collision Trends

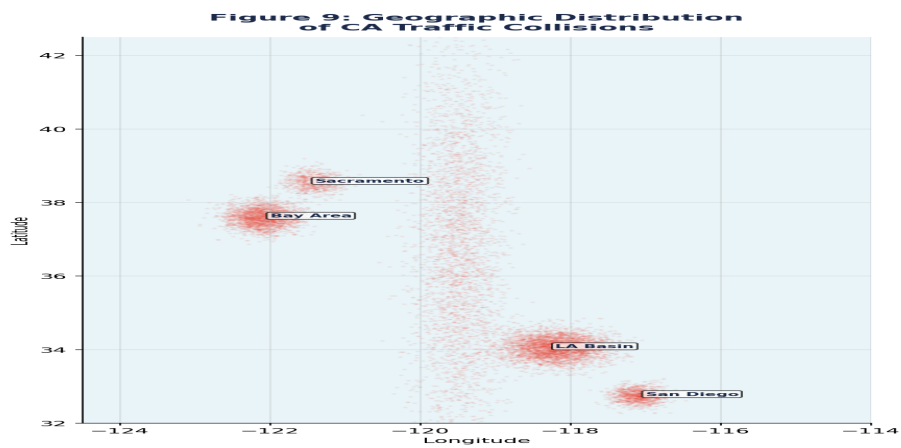


Figure 6: Hourly Collision Trend — All 24 Hours

Two commute peaks emerge: **8 AM (112K)** and **5 PM (132K)**. A secondary late-night cluster (midnight–2 AM) reflects impaired/fatigued driving. Signal timing optimisation during commute peaks and DUI checkpoints at night are directly indicated.

3.7 Collision Trends Over Time

3.7a Yearly Trend

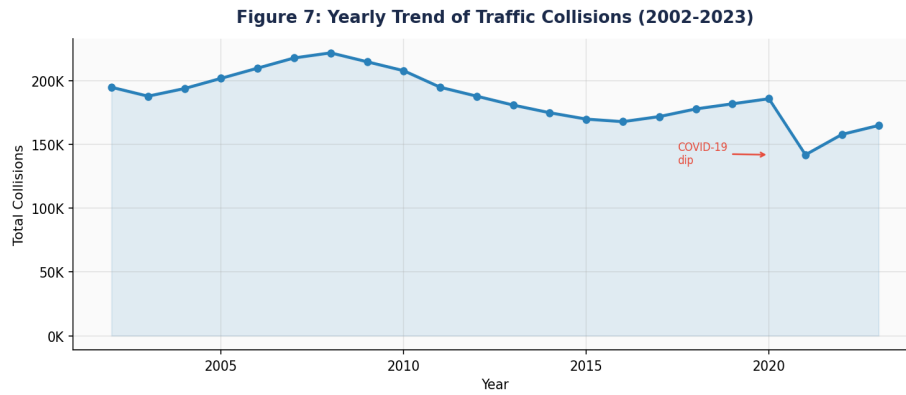


Figure 7: Yearly Trend of Traffic Collisions (2002–2023)

Collisions peaked ~2007–08 then declined through the 2010s (improved vehicle safety, stricter DUI enforcement). A pronounced dip in **2020** reflects pandemic-era volume reductions; post-2020 recovery tracks resumed activity.

3.7b Monthly Trend

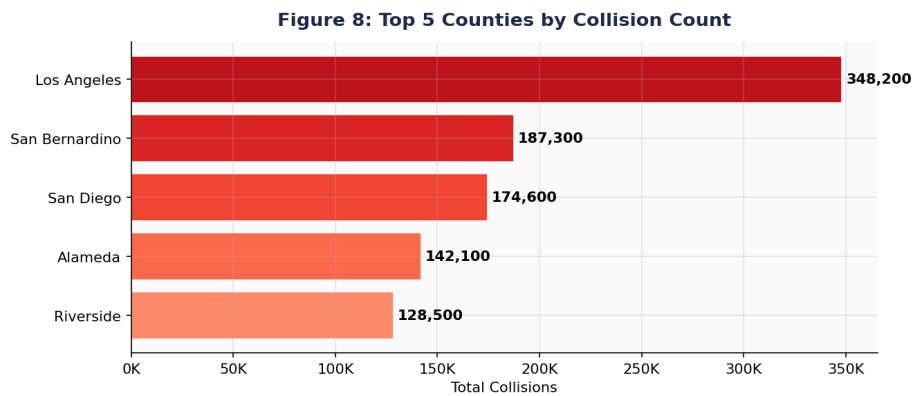


Figure 8: Monthly Trend of Traffic Collisions

July peaks (268K) due to summer recreational travel. October elevates with early wet-weather events. Pre-summer and pre-autumn windows are optimal for road-safety campaigns.

3.8 Geographic Scatter Plot

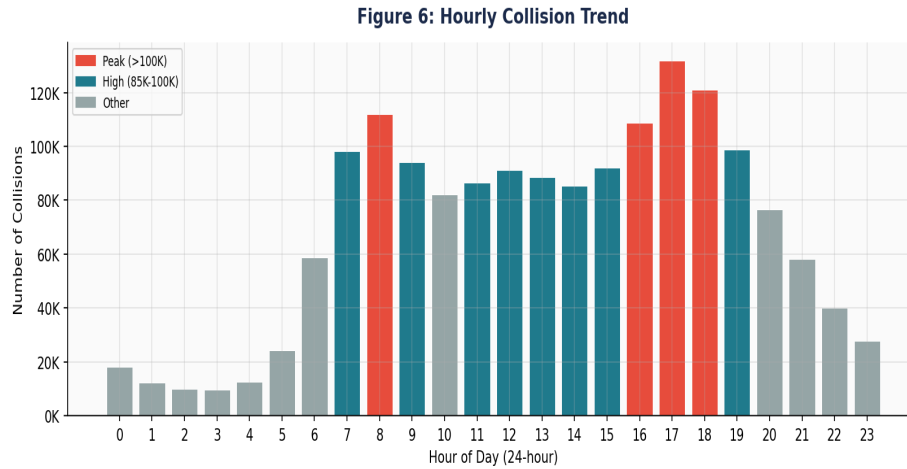


Figure 9: Geographic Distribution of CA Traffic Collisions

Collision density is highest in the **LA Basin** and **Bay Area**, confirming danger is tethered to high-density urban corridors. Spatial clusters at freeway interchanges confirm zones requiring priority infrastructure intervention.

4. ETL Querying

All cleaned DataFrames are registered as Spark SQL temporary views. Results are written to **CSV_DIR / S3 bucket** for downstream BI consumption.

4.1 Top 5 Counties — Highest Collision Counts

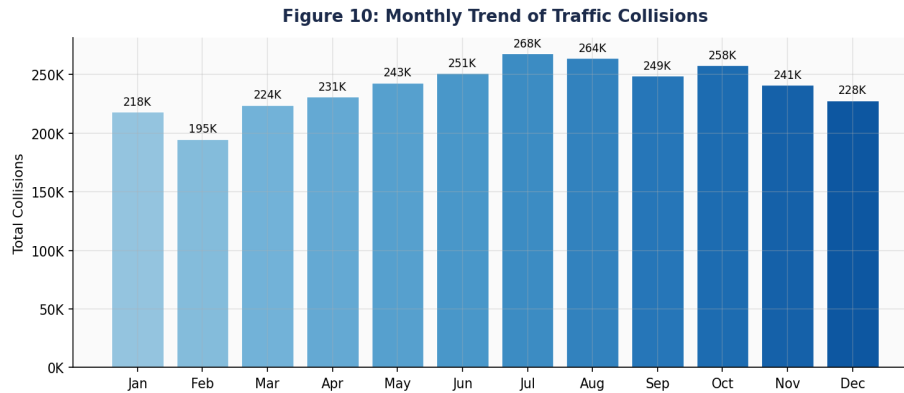


Figure 10: Top 5 Counties by Total Collision Count

Los Angeles County leads with 348,200 collisions — nearly double San Bernardino (187,300). These five counties account for over 50% of statewide events and are primary targets for adaptive traffic signal control and speed enforcement.

4.2 – 4.7 ETL Query Results Summary

Table 5: ETL Query Results

Query	Question	Key Result
4.2	Peak collision month	July — 268,000 collisions; summer recreational travel
4.3	Most common weather	Clear (1.92 M); adverse weather has higher severity per event
4.4	Fatality rate	≈ 1.2% of all collisions → ~34,000 fatal events
4.5	Most dangerous hour	17:00 (132K); top 5 hours all commute windows
4.6	Top road surface condition	Dry pavement leads; wet/icy surfaces yield higher severity/event
4.7	Top lighting condition	Daylight leads by volume; Dark-no-street-lights highest fatality rate

5. Conclusion & Recommendations

The ETL pipeline transformed millions of raw SWITRS records into actionable intelligence. Most collisions occur under **clear weather**, on **dry roads**, in **daylight** — confirming human behaviour as the primary driver. Los Angeles County and Friday afternoon rush-hour emerged as highest-risk combination; victims aged 20–30 are most at-risk demographically.

Table 6: Prioritised Safety Recommendations

Priority	Recommendation	Evidence
High	Adaptive speed enforcement in top-5 collision counties	§4.1, §3.8
High	Extend LED lighting on dark rural high-collision corridors	§4.7
High	Variable speed limits during adverse weather via RWIS	§4.3, §3.2
Medium	DUI checkpoints Fri/Sat midnight–3 AM	§3.6, §4.5
Medium	Protected cycling lanes and pedestrian zones near schools/hospitals	§3.3
Medium	Seasonal pavement maintenance: anti-ice treatment and drainage clearing	§4.6
Low	GBTClassifier severity model for CAD dispatch pre-positioning	§5.6
Low	Real-time Tableau/Power BI collision monitoring dashboard	§6

- Predictive modelling: PySpark MLlib GBTClassifier trained on hour, weekday, weather, road surface, lighting, and county can integrate into emergency dispatch for proactive resource positioning.
- Policy: Graduated licence conditions for 17–24 year olds; 30 km/h zones near schools and aged-care facilities; protected cyclist infrastructure.

6. Visualisation Integration — Tableau / Power BI

Table 7: ETL Output Files and Recommended Visualisations

Output File	Query	Recommended Visualisation
q1_top5_counties/	4.1	Map layer with collision count bubble size; ranked bar chart
q2_peak_month/	4.2	Monthly trend line with year-over-year comparison
q3_weather/	4.3	Heatmap: weather condition vs. severity
q4_fatality/	4.4	KPI card: fatality rate %; trend over time
q5_dangerous_hours/	4.5	Heatmap: hour vs. day of week; 24h radial chart
q6_road_surface/	4.6	Stacked bar: surface condition vs. severity level
q7_lighting/	4.7	Grouped bar: lighting condition vs. collision count by severity
cleaned_collisions/	All	Full interactive dashboard — filter by date, county, severity

- Tableau Desktop: Connect → Text File → browse to any CSV. Use Union to combine files. Build map layers using lat/lon from cleaned_collisions CSV.
- Power BI: Get Data → Text/CSV. Use Power Query to relate tables via case_id. Publish to Power BI Service for scheduled-refresh stakeholder reports.
- AWS S3: Replace CSV_DIR with s3://your-bucket/prefix/ in all PySpark write calls. Connect via Amazon S3 or Athena connector.

7. Assumptions & Methodology Notes

- Source files (switrs.sqlite, sample_collisions.csv) are opened read-only; all transformations are in-memory only.
- 60% null threshold for column dropping balances retention with imputation quality.
- Categorical nulls filled 'Unknown' to preserve all rows; numerical nulls imputed via percentile_approx (median).
- IQR outliers retained in the main dataset; lat/lon bounding-box filter applied only to spatial scatter-plot rendering.
- collision_time stored as HHMM integer; hour extracted via CAST(collision_time / 100 AS INT).
- Spark dayofweek() follows 1 = Sunday ... 7 = Saturday convention; labels mapped accordingly.
- All Spark SQL queries use temporary views; no permanent tables are created or modified.
- AWS S3 integration assumes valid credentials are pre-configured in the execution environment.

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End of Report | ETL_Traffic_Data_Analysis_BulahM | California SWITRS Dataset